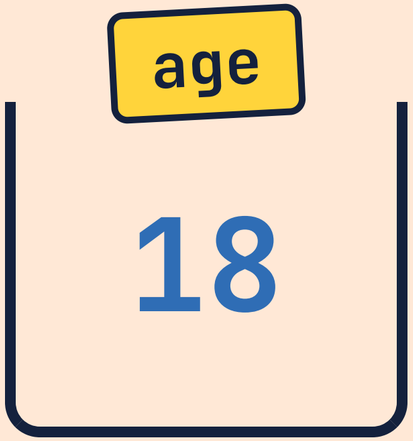
CODE IN SMALL STEPS · LESSON 1

Variables: labelled boxes for values

Python basics · 5 minutes · one tiny exercise



A STRANGE RESULT

Same symbol. Different answer.

```
5 + 2
```

7

```
"5" + "2"
```

"52"

Why? By the end of this lesson, you'll know.

IN THIS LESSON

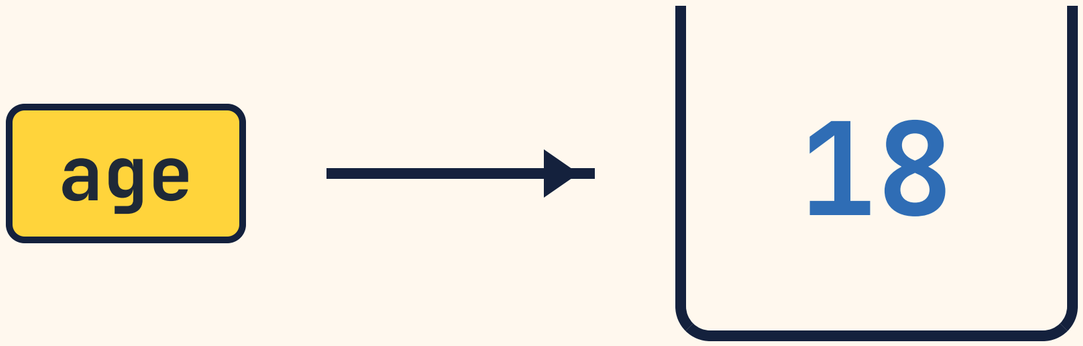
You will be able to...

1. Create a variable and give it a value
2. Tell `int` , `float` and `str` apart
3. Predict what `+` does with each type
4. Fix the most common variable error

THE IDEA

A variable is a name that points to a value

```
age = 18
```



The = sign means “store”, not “is equal to”.

THREE BASIC TYPES

Every value has a type

int

```
age = 18
```

whole numbers

float

```
price = 2.5
```

numbers with a decimal point

str

```
name = "Mina"
```

text, always inside quotes

Check any value with `type(…)`

WHAT + MEANS

The type decides what + does

int + int

5 + 2 → 7

adds the numbers

str + str

"5" + "2" → "52"

joins the text

"5" + 2 → TypeError: Python won't guess which one you meant

CONVERTING

Change the type on purpose

```
int("5") + int("2")    # 7
str(18) + " years"     # "18 years"
float("2.5") * 2       # 5.0
```

Text from `input()` is always a `str`, so convert it before doing maths.

YOUR TURN

Pause and try it

1. Store two numbers in two variables
2. Print their sum
3. Bonus: print it inside a sentence



MOST COMMON ERROR

Names must match exactly

```
score = 40
print(scor)

Traceback (most recent call last):
  File "lesson.py", line 2, in <module>
    print(scor)
          ^^^^
NameError: name 'scor' is not defined. Did you mean: 'score'?
```

Read the **last line** first. Python even suggests the fix.

RECAP

Three things to remember

1. A variable is a name that points to a value
2. The type changes what operators like `+` do
3. Convert types on purpose: `int()` , `str()` , `float()`

NEXT LESSON If / else: let your program choose